

Project 62: Text-to-Speech with ElevenLabs or Coqui

Goal: Turn any input text into **audio speech output**, with support for voice selection, saving to file, and real-time playback.

Two Implementation Options

Platform

Pros

Cons

ElevenLabs

Ultra-realistic voices, easy API

Requires API key, cloud

Coqui TTS

Offline, open-source, customizable

Heavier install

OPTION 1: ElevenLabs TTS (Cloud-Based)

Step 1: Install & Set API Key

```
pip install elevenlabs
```

```
from elevenlabs import generate, play, save, set_api_key

set_api_key("your-elevenlabs-api-key")
```

Step 2: Convert Text to Speech

```
text = "Welcome to the AI Text-to-Speech demo using ElevenLabs!"
audio = generate(text=text, voice="Rachel", model="eleven_monolingual_v1")

# Playback
play(audio)

# Save to file
save(audio, "output_elevenlabs.wav")
```

ElevenLabs Summary

- **Input:** Text
 - **Voice:** Predefined (e.g., Rachel, Adam, custom)
 - **Output:** WAV file or live playback
 - **Use Case:** Voiceovers, AI narration, content accessibility
-

OPTION 2: Coqui TTS (Fully Offline)

Step 1: Install Coqui

```
pip install TTS
```

Optionally download models: `tts --list_models`

Step 2: Load and Use a Pretrained TTS Model

```
from TTS.api import TTS

# Load multilingual or English model
tts = TTS(model_name="tts_models/en/ljspeech/tacotron2-DDC", progress_bar=False)

# Synthesize speech and save
tts.tts_to_file(text="Hello! This is an offline voice using Coqui TTS.", file_
```

Coqui TTS Summary

- **Offline**
 - Many model options (Tacotron, FastSpeech, GlowTTS, etc.)
 - Languages: English, German, Spanish, etc.
 - Can be fine-tuned with your own data
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Project Summary

Feature

ElevenLabs

Coqui TTS

Online/Offline

Online (API)

Offline (local)

Voices

Hyper-realistic, cloned voices

Basic to advanced TTS models

Customization

High (premium features)

High (train your own)

Ideal for

Demos, apps, realistic speech

Private, offline, research use